

GURMOHIT SINGH

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EDUCATION

The University of Texas at Austin - GPA: 3.95

Bachelor of Science, Computer Science

Aug 2024 – May 2028 (Expected)

Lone Star Community College - GPA: 3.98

Associate of Science

Aug 2021 – May 2024

SKILLS

Programming Languages: Python, C, C++, Java, TypeScript, JavaScript, C#

ML/Data: pandas, NumPy, scikit-learn, LightGBM, Matplotlib, TensorFlow, PyTorch, feature engineering, cross-validation, hyperparameter tuning, reinforcement learning, Bayesian networks, neural networks

Systems/Tools: Git, Linux, ARM Assembly, GPU-Z telemetry, performance benchmarking, Qiskit, IBM Quantum, PySide6, Excel

Web Development: React, Next.js, Vite, Tailwind CSS, Spring Boot, HTML/CSS, Vercel, Cloudflare Pages

Coursework: Machine Learning, Artificial Intelligence, Operating Systems, Computer Architecture, Quantum Computing, Energy-Efficient Computing, Algorithms

EXPERIENCE

WPSPIN LLC – Houston, TX

Web Development Intern

Jun – Aug 2026

- Developed static and WordPress-based websites for local business clients, translating design/content requirements into responsive, production-ready pages.
- Leveraged agentic workflows to enhance productivity, ~50% faster project turnover time and improved codebase maintainability.
- Used HTML, CSS, and JavaScript tooling to improve page structure, mobile responsiveness, and maintainability across client sites.

Tandoorish LLC – Houston, TX

IT Services Manager

Jun 2024 – Present

- Managed business-critical technology systems including POS workflows, online ordering, website updates, email/account access, printers, and Wi-Fi/network troubleshooting.
- Served as the first point of contact for staff technology issues, resolving platform access, device, ordering, connectivity, and vendor-support problems during daily operations.
- Used SEO and Google Analytics insights to improve online visibility, contributing to a 200%+ increase in site traffic since takeover.

Operations Manager

Nov 2022 – Present

- Led daily operations, staff training, inventory management, and customer service in a fast-paced, high-volume environment.
- Assisted with food preparation, kitchen coordination, quality control, and menu execution to ensure consistency and efficiency.
- Improved cost control through supplier coordination, budgeting, inventory oversight, and waste reduction initiatives, lowering operational costs by approximately 30%.

PROJECTS

[Aviation Damage Risk Predictor](#) | Python, Pandas, scikit-learn, LightGBM, NumPy

Mar – May 2026

- Built an imbalanced binary classification pipeline to predict indicated aircraft damage from 307K+ aviation incident records.
- Engineered 104 interpretable features from missingness patterns, date/time fields, high-cardinality categorical variables, text keywords, frequency encodings, and domain-specific risk interactions.
- Evaluated LightGBM variants with 5-fold stratified cross-validation, threshold tuning, and ablation testing; achieved 0.939 out-of-fold ROC-AUC and 0.860 balanced accuracy.

[Pacman AI Project Suite](#) | Python, Search, Reinforcement Learning, Bayesian Networks

Jan – May 2026

- Implemented AI agents across five Pacman projects covering graph search, adversarial search, value iteration, Q-learning, Bayesian network inference, and supervised classification.
- Applied these methods to maze navigation, multi-agent decision-making, reinforcement learning, probabilistic reasoning, and digit/Pacman behavior classification.

[ARM Visualizer](#) | TypeScript, React, Vite, Vitest, Playwright

Dec 2025 – Present

- Building an interactive AArch64 simulator with a five-stage, in-order pipeline and browser-based assembly execution controls.
- Implementing pipeline state visualization, register/memory inspectors, cycle history, instruction focus views, forwarding paths, bubbles, stalls, flushes, and retirement tracking.
- Added unit, React, and Playwright end-to-end tests covering CPU execution, pipeline hazards, forwarding, memory behavior, invalid programs, and UI workflows.

[Energy-Efficient Graphics API Benchmark](#) | Vulkan, Direct3D 12, OpenGL, GPU-Z

Feb – May 2026

- Conducted a controlled GPU efficiency study across Vulkan, Direct3D 12, and OpenGL APIs with various graphical workloads.
- Collected benchmark scores, FPS, and power/thermal metrics across repeated trials, then derived performance-per-watt metrics.
- Found Vulkan delivered the best measured performance per watt in both benchmarks, including 0.727 FPS/W in GravityMark, versus 0.641 FPS/W for OpenGL and 0.633 FPS/W for Direct3D 12, at roughly equal board power.

ACTIVITIES

Summer 2025 Quantum Optics Program - Applied Research Laboratories, UT Austin – Austin, TX

Jun 2025 – Aug 2025

- Gained hands-on experience with quantum computing algorithms and their implementations with optical lab equipment.
- Implemented algorithms such as Quantum Teleportation and Entanglement Swapping and demonstrated a violation of the Bell-CHSH Inequality